

# Automatic Control of Aircraft, AE40019

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**Project:** *Simulation of the paper, “Analysis and Design of Launch Vehicle Flight Control Systems, by Bong Wie and Wei Du, Iowa State University, Ames, IA 50011-2271 and Mark Whorton, NASA Marshall Space Flight Center, Huntsville, AL 35812”*

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## Introduction

The paper chosen describes the fundamental principle of launch vehicle flight control analysis and design. In particular, designing the control system for pitch angle control with particular focus on the classical concept of “drift-minimum” and “load minimum” control principles.

## Theory

Considering a simplified linear dynamical model of a launch vehicle. As illustrated in figure below, as follows:

$$\begin{aligned}\ddot{\theta} &= M_{\alpha}\alpha + M_{\delta}\delta \\ \ddot{Z} &= -\frac{F}{m}\theta - \frac{N_{\alpha}}{m}\alpha + \frac{T}{m}\delta \\ \alpha &= \theta + \frac{\dot{Z}}{V} + \alpha_w \\ F &= T_0 + T - D\end{aligned}$$

Where,  $\theta$  is the pitch attitude,  $\alpha$  the angle of attack,  $Z$  the inertial Z-axis drift position of the center-of-mass,  $\dot{Z}$  the inertial drift velocity,  $m$  the vehicle mass,  $T_0$  the un-gimbled sustainer thrust,  $T$  the gimbaled thrust,  $N = N_{\alpha}\alpha$  the aerodynamic normal (lift) force acting on the center-of-pressure,  $D$  the aerodynamic axial (drag) force,  $F$  the total x-axis force,  $\delta$  the gimbal deflection angle,  $V$  the vehicle velocity,  $\alpha_w = \frac{V_w}{V}$  the wind-induced angle of attack,  $V_w$  the wind disturbance velocity, and

$$\begin{aligned}M_{\alpha} &= \frac{N_{\alpha}}{I_y}x_{cp} \\ M_{\delta} &= \frac{T}{I_y}x_{cg}\end{aligned}$$

$$\frac{\dot{Z}(s)}{\alpha_w(s)V} = -\frac{A_2s^2 + A_1s + A_0}{s^3 + B_2s^2 + B_1s + B_0}$$

$$\frac{\alpha(s)}{\alpha_w(s)} = \frac{s(s^2 + M_\delta K_2 s + M_\delta K_1)}{s^3 + B_2 s^2 + B_1 s + B_0}$$

$$\frac{\delta(s)}{\alpha_w(s)} = \frac{s(K_3 s^2 + M_\alpha K_2 s + M_\alpha K_1)}{s^3 + B_2 s^2 + B_1 s + B_0}$$

where

$$B_2 = M_\delta K_2 + \frac{T}{mV} \left( K_3 + \frac{N_\alpha}{T} \right)$$

$$B_1 = M_\delta (K_1 + K_3) - M_\alpha + \frac{K_2 T}{mV} \left( M_\alpha + \frac{M_\delta N_\alpha}{T} \right)$$

$$B_0 = \frac{T K_1}{mV} \left( M_\alpha + \frac{M_\delta N_\alpha}{T} \right) - \frac{F}{mV} (M_\delta K_3 - M_\alpha)$$

$$A_2 = \frac{T}{mV} \left( K_3 + \frac{N_\alpha}{T} \right)$$

$$A_1 = \frac{K_2 T}{mV} \left( M_\alpha + \frac{M_\delta N_\alpha}{T} \right)$$

$$A_0 = B_0$$

Now finding the steady state value of the drift velocity, when the wind disturbance is a constant step function  $\alpha_w(s) = 1/s$

$$\frac{\dot{Z}_{ss}}{V} = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times -\frac{A_2 s^2 + A_1 s + A_0}{s^3 + B_2 s^2 + B_1 s + B_0} = -\frac{A_0}{B_0} = -1$$

$$\Rightarrow \dot{Z}_{ss} = -V$$

Since, we can notice that the steady state drift velocity does not depend on the gains set by us, that means we can not control the steady state drift velocity, but if we make  $B_0 = 0$ , steady state value become,

$$\frac{\dot{Z}_{ss}}{V} = -\frac{A_1}{B_1} = -\frac{1}{1+C}$$

Where

$$C = \frac{mV[M_\delta(K_1+K_3)-M_\alpha]}{M_\alpha K_2 T + M_\delta N_\alpha / T}$$

Now for, stable closed loop system, the thrusters must have the necessary control to negate the moments generated by the aerodynamic forces, this implies,  $M_\delta(K_1 + K_3) - M_\alpha > 0$ , we have  $C > 1$ , implies  $|\dot{Z}_{ss}| < V$ , since the steady state values of  $\delta = \alpha = 0$ , then the maximum lateral velocity can be the wind disturbance,  $V_w$ , hence  $|\dot{Z}_{ss}| < V_w$ .

Hence, the condition,  $B_0 = 0$  is the *Drift minimum condition*, can be rewritten as,

$$\frac{M_\delta K_3 - M_\alpha}{M_\delta K_1} = \frac{N_\alpha}{F} \left( 1 + \frac{x_{cp}}{x_{cg}} \right)$$

Now for load minimum case, since we know that total moment acting on the launch vehicle is the sum of the effects due to angle of attack,  $\alpha$  and gimbal angle  $\delta$ . Hence upon minimizing the steady state values of  $\alpha$  and  $\delta$  will give the load minimum condition.

Since, steady state values,

$$\alpha_{ss} = M_\delta K_1 \text{ and } \delta_{ss} = M_\alpha K_1$$

From the above expressions if we see that, upon putting  $K_1 = 0$  will give the minimum steady state angle of attack and steady state gimbal angle. This is the condition for **Load minimum control** of the launch vehicle.

### Analysis and Design

For the vehicle considered in the paper, with specifications,

$$I_y = 2.43E6 \text{ slug} - ft^2, \quad m = 5830 \text{ slug}, \quad T = 341,000 \text{ lb},$$

$$F = 375,000 \text{ lb}, \quad x_{cp} = 38 \text{ ft}, \quad x_{cg} = 32.3 \text{ ft}, \quad V = 1320 \frac{ft}{sec},$$

$$V_w = 132 \frac{ft}{sec}, \quad \alpha_w = 5.73 \text{ deg}, \quad N_\alpha = 240,000 \frac{lb}{rad}, \quad M_\alpha = 3.75 s^{-2},$$

$$M_\delta = 4.54 s^{-2}$$

And gimbal actuator has 2<sup>nd</sup> order dynamics,

$$\delta(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \delta_c(s)$$

This implies that after setting the gains using the pole placement method the control input goes from the actuator dynamics and serves to the actual system.

The plant

$$\dot{X} = AX + B\delta$$

$$Y = CX$$

For observing all the states,  $C = eye(3)$

$$X = \begin{bmatrix} \theta \\ \dot{\theta} \\ \alpha \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & M_\alpha \\ -F/mV & 1 & -N_\alpha/mV \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ M_\delta \\ T/mV \end{bmatrix}$$

Since gimble actuator has 2<sup>nd</sup> order response, then gimble system,

$$\dot{\bar{X}} = \bar{A}\bar{X} + \bar{B}\delta_c$$

Since,  $\delta_c = -KX$ , then,

$$\dot{\bar{X}} = \bar{A}\bar{X} - \bar{B}KX$$

And,

$$\delta = \bar{C}\bar{X}$$

Where,

$$\bar{X} = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 0 & 1 \\ -2500 & -100 \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} 0 \\ 2500 \end{bmatrix}, \quad \bar{C} = [1 \quad 0]$$

Putting the value of  $\delta$  in plant equation, the overall equation become,

$$\dot{X} = AX + B\bar{C}\bar{X}$$

$$\dot{\bar{X}} = \bar{A}\bar{X} - \bar{B}KX$$

$$Y = CX$$

Since we also want to get the values of drift and drift velocity,

$$Adrift = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1; -(F/m) \ 0 \ -(N\_alpha/m) \ (T/m) \ 0 \ 0 \ 0];$$

now combining these into one system equation,

$$\dot{\tilde{X}} = \tilde{A}\tilde{X}$$

$$\tilde{Y} = \tilde{C}\tilde{X}$$

Where,

$$\tilde{X} = \begin{bmatrix} \theta \\ \dot{\theta} \\ \alpha \\ \delta \\ \dot{\delta} \\ Z \\ \dot{Z} \end{bmatrix}$$

$$\tilde{A} = [A, B*\tilde{C}, zeros([3,2]); -\tilde{B}*K, Atilde, zeros([2,2]); Adrift];$$

## Results

| Case No. | Controller Type                                 | Feedback Gains ( $K_1, K_2, K_3$ ) | Closed-Loop Poles              |
|----------|-------------------------------------------------|------------------------------------|--------------------------------|
| 1        | $(\theta, \dot{\theta})$ -Feedback Control [15] | (2, 0.8, 0)                        | $-1.7488 \pm 1.3934j, -0.1596$ |
| 2        | Drift-Minimum Control [15]                      | (2, 0.8, 3.614)                    | $-1.9087 \pm 4.2774j, 0.0$     |
| 3        | Load-Minimum Control [15]                       | (0, 0.8, 3.614)                    | $-1.9323 \pm 3.0533j, 0.0471$  |
| 4        | LQR Control ( $Q = 0$ )                         | (0.6852, 0.8491, 0.9542)           | $-1.9767, -1.8967, -0.0488$    |
| 5        | Drift-Minimum Control                           | (0.3220, 0.8352, 1.2765)           | $-1.9767, -1.8967, 0.0$        |
| 6        | Load-Minimum Control                            | (0, 0.8352, 1.2765)                | $-3.1323, -0.7816, 0.0405$     |

### 1. Feedback Control

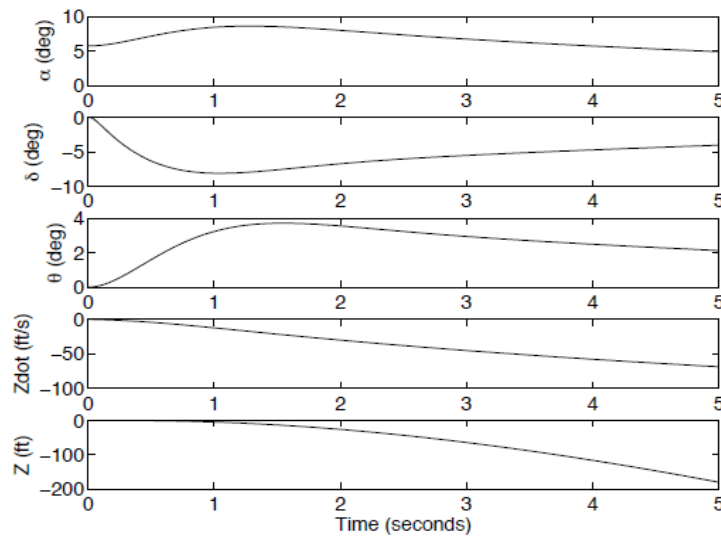


Figure 3.  $(\theta, \dot{\theta})$ -feedback control (Case 1).

Figure 2 Parameter variations given in the Paper for Feedback Control

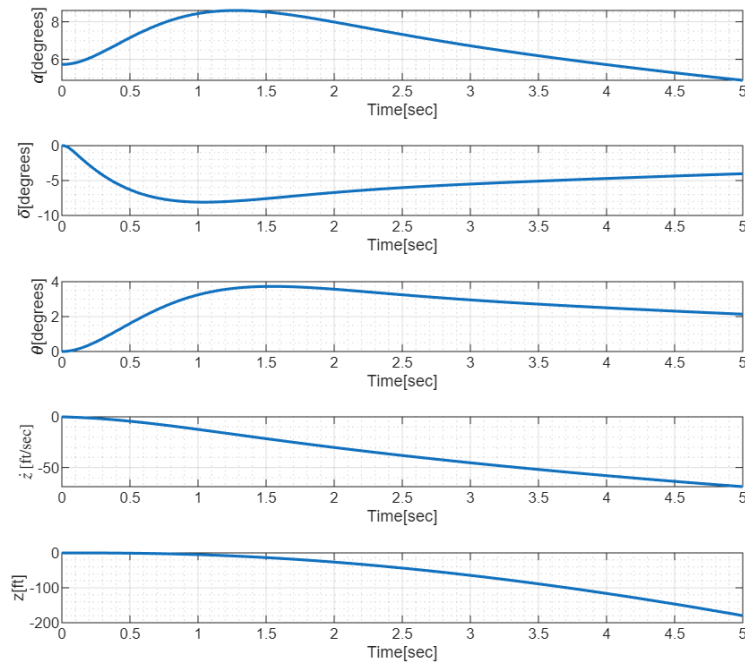


Figure 3 Simulate Parameter Variation for Feedback Control

## 2. Drift - Minimum Control

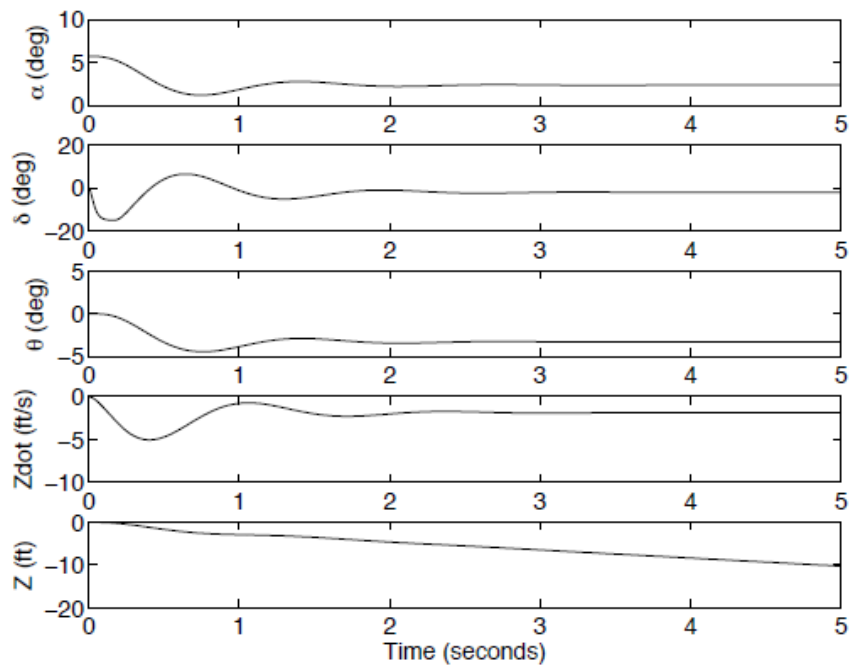


Figure 4. Drift-minimum control (Case 2).

Figure 4 Parameter Variation Given in the Paper for Drift Minimum Control

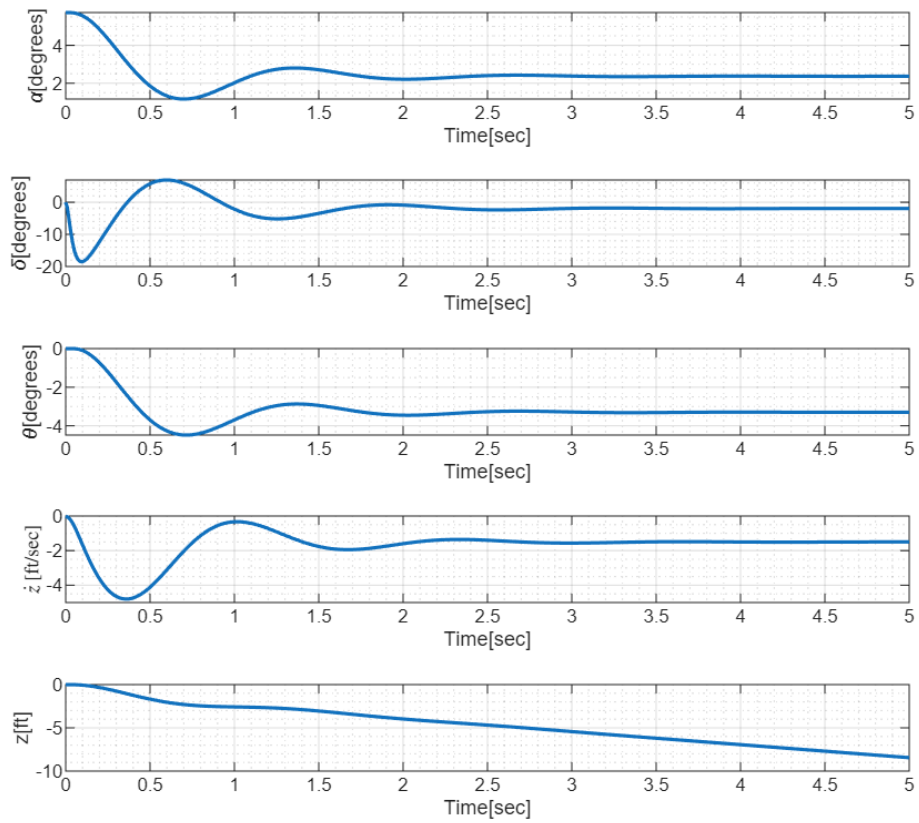


Figure 5 Simulated Parameter Variations for Drift Minimum Control

### 3. Load – Minimum Control

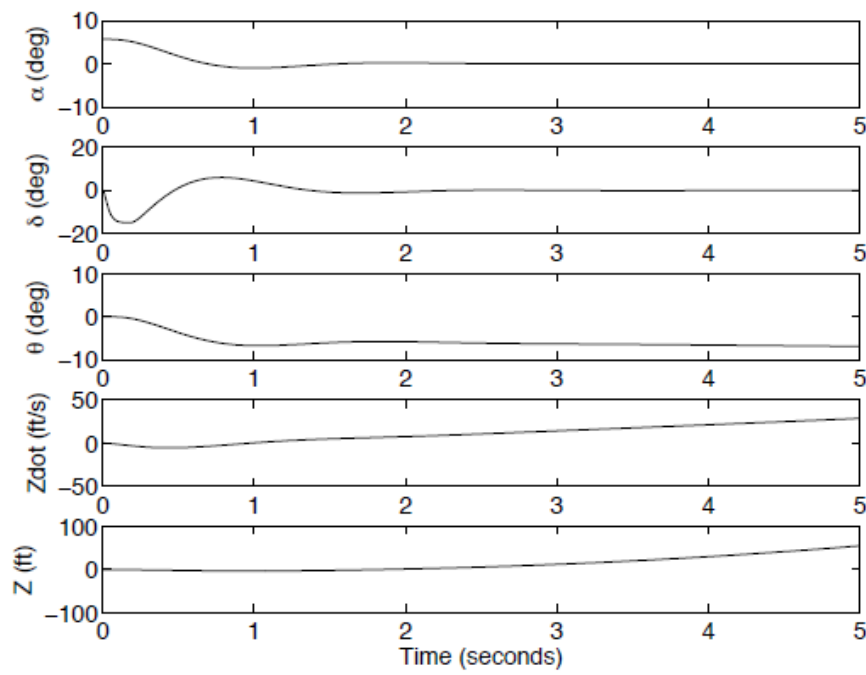


Figure 5. Load-minimum control (Case 3).

Figure 6 Parameter Variation Given in the Paper for Load Minimum Control

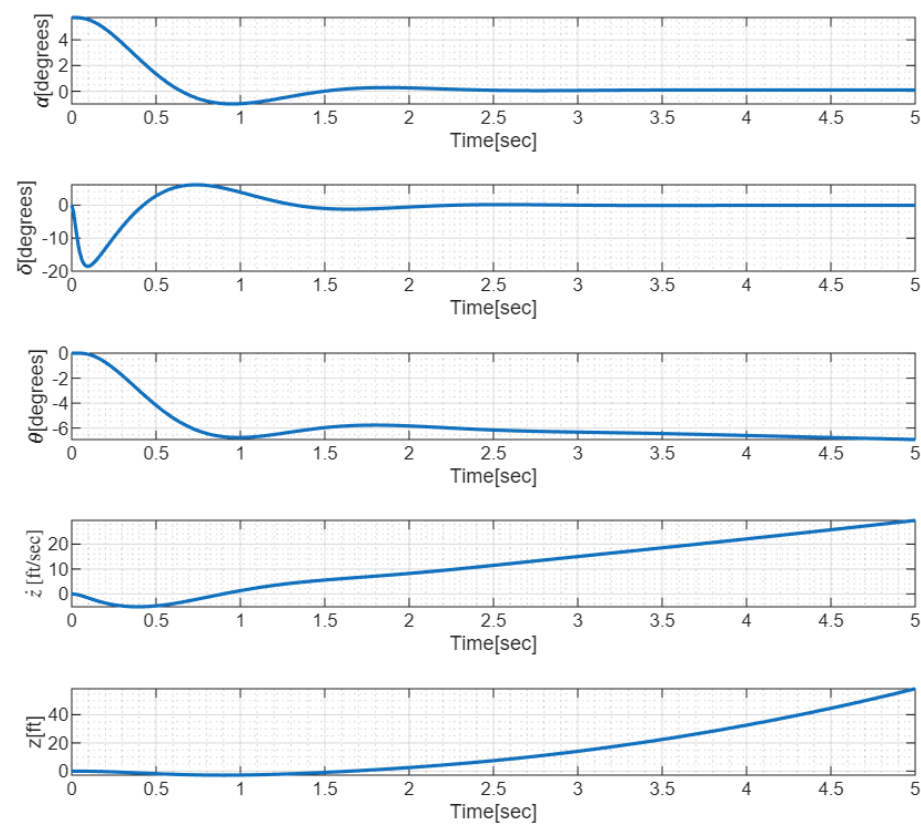


Figure 7 Simulated Parameter Variations for Load Minimum Control

4. Linear Quadratic Regulator Control (LQR)

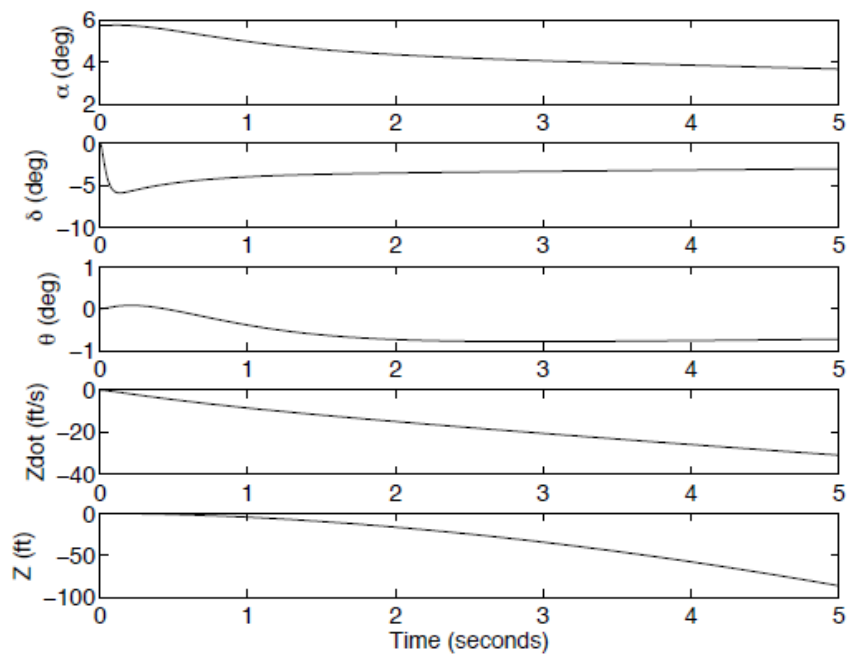


Figure 6. LQR control (Case 4).

Figure 8 Parameter Variation Given in the Paper for LQR Control

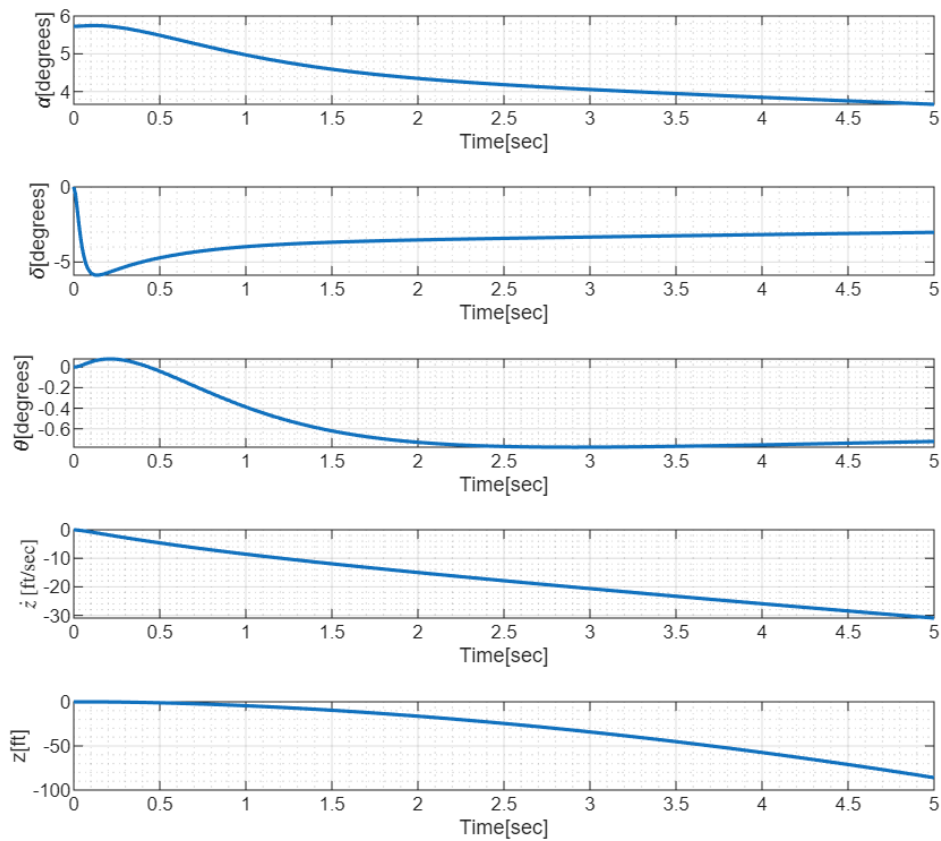


Figure 9 Simulated Parameter Variations for LQR Control

## 5. Drift Minimum Control with Different Desired Closed-Loop Poles

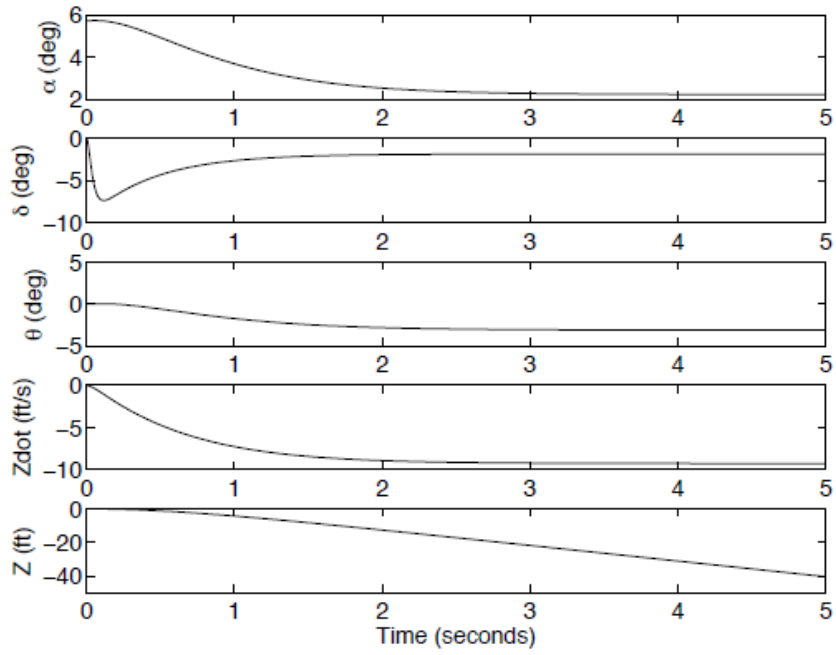


Figure 7. Drift-minimum control (Case 5).

Figure 10 Parameter Variation Given in the Paper for Drift Minimum Control with different Closed-Loop Poles

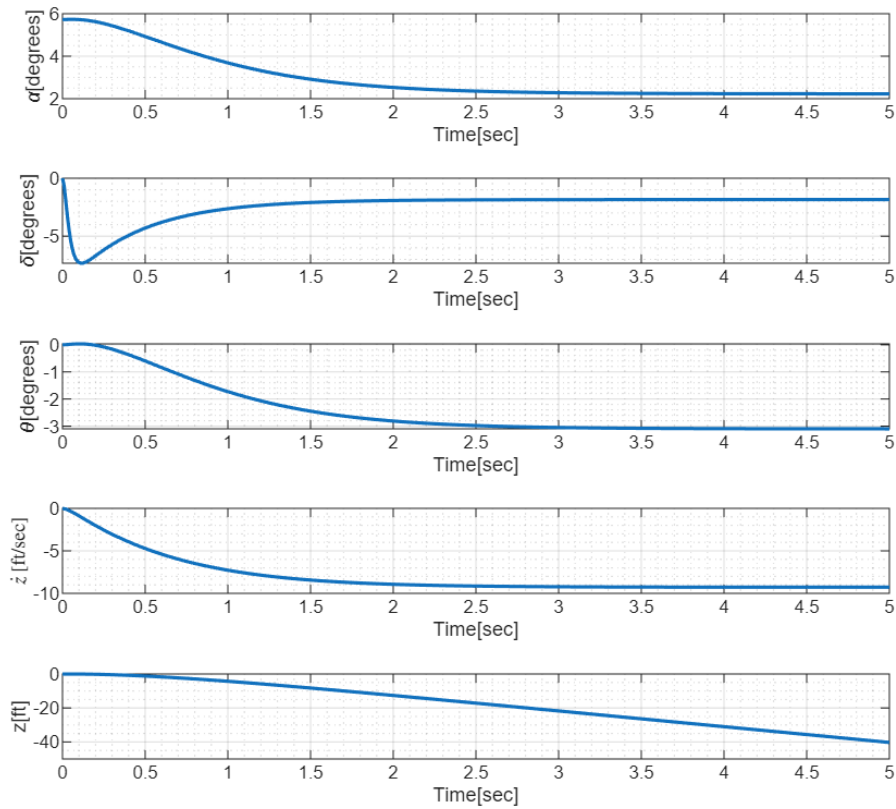


Figure 11 Simulated Parameter Variations for Drift Minimum Control with Different Closed-Loop Poles

## 6. Load Minimum Control with different Closed-Loop Poles

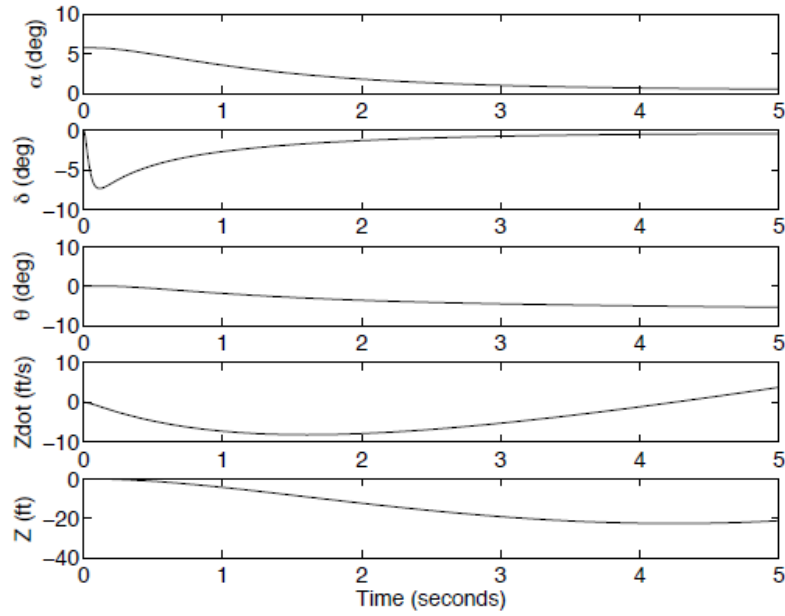


Figure 8. Load-minimum control (Case 6).

Figure 12 Parameter variations given in Paper for Load Minimum Control with different Closed-Loop Poles

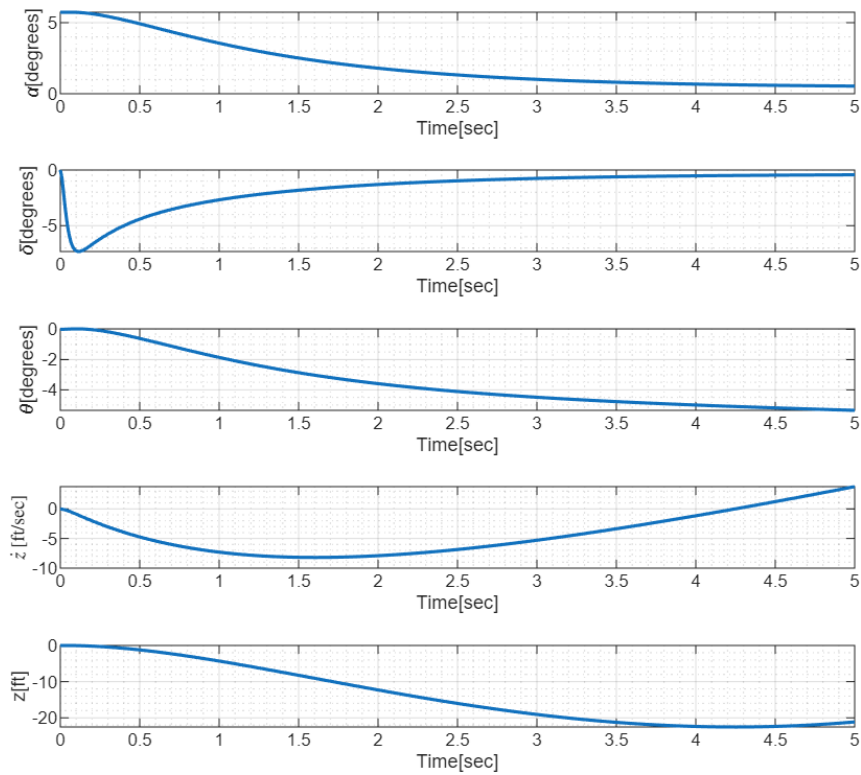


Figure 13 Simulated Parameter Variations for Load Minimum Control with Different Closed-Loop Poles

## Appendix

### Simulink block diagram of the system

